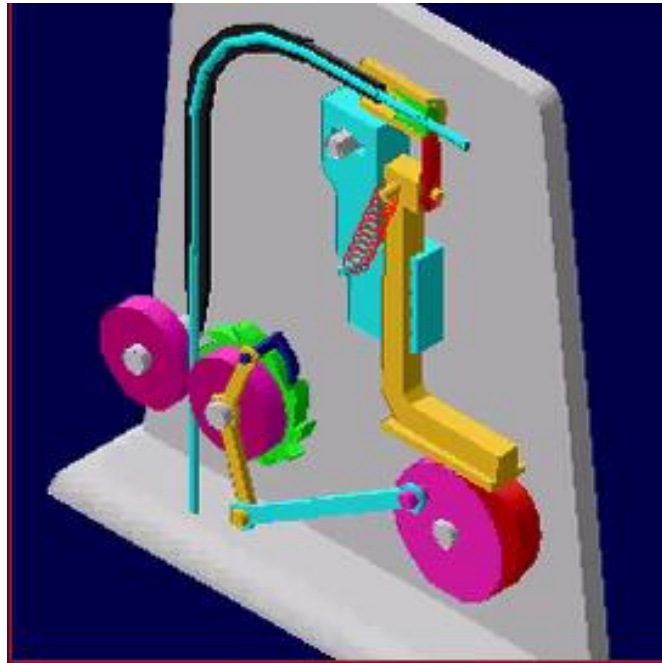
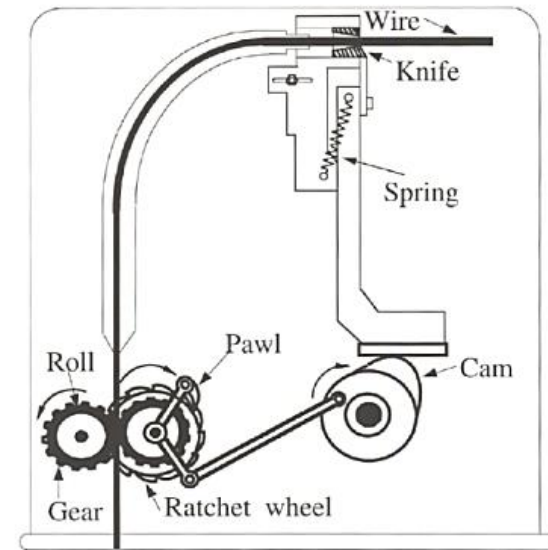


Sophisticated and complex mechanical systems are composed of simple mechanisms.



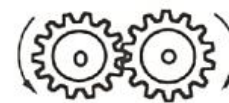
Wire feeding and cutting mechanism



Cam-Follower



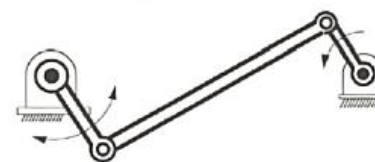
Gear pair



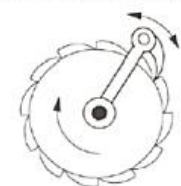
Roll pair



Crank-Rocker



Pawl & Ratchet wheel



“Nothing is particularly hard if you divide it into small jobs.”

-- Henry Ford

Course Overview and Logistics

ME 350 Lecture #1

Agenda

- Introductions of Instructional Team Members
- Machine shop training
- Logistics of the course
- Project
- Mechanisms

Mike Umbriac

Background:

- Worked for Ford Motor Company as a Manufacturing Engineer for Engine Assembly for 17 years
- Left Ford to come back to U of M as a Lecturer.



Fun Facts:

- Likes running and skiing
- Was a GSI for ME350 in 1991 and 1992
- His son was born in the car

he/him/his



Prof. Diann Brei

Background

- Professor of Mechanical Engineering
(1994-present, UG Chair 2012-2017, Chair of ISD 2018-2023)
- Research in Smart Materials and Structures; Design Science
- First woman to win the ASME Machine Design Award

Fun Facts

- I love to Design and take things apart
(be careful – I break everything I touch!)
- Developed and taught design curriculum for over 30 years
- Big family – 4 kids (3 at UM) – we are our own party!
- Former Girl Scout Leader
- First Robotic Mom



Jeffrey Koller

Background

- Ph.D. & M.S. in Mechanical Engineering from the University of Michigan
 - Research on adaptive controls for exoskeletons and prostheses
- B.S in Mechanical Engineering from the University of California, Berkeley
- Was a technical consultant with Exponent for a year after graduate school (really didn't like it...)
- Extremely happy to be in a teaching track faculty position and have been teaching for ~7 years now (Villanova & Michigan)



Fun Facts

- Sit in the same office I interviewed for graduate school in
- I have a pet lorax named Oliver
- Spend my spare time with family, biking, running, cooking, and reading



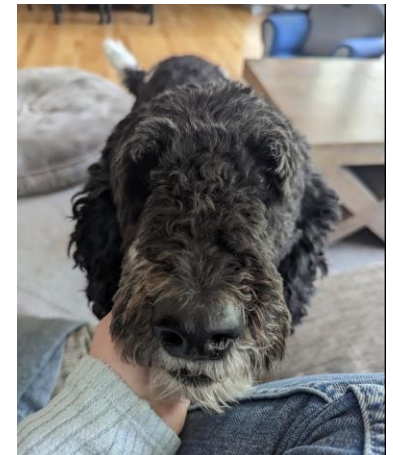
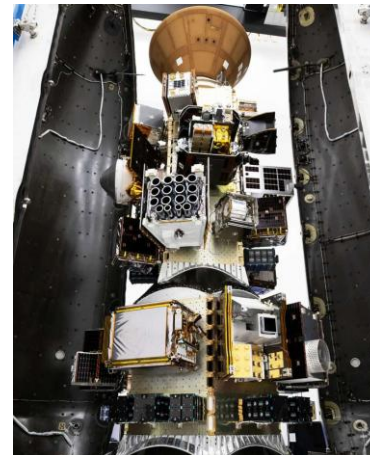
Chloe Affaki

Background

- SUGS student in Mechanical Engineering
- 5th semester as a lab instructor for ME 350
- Interned at SpaceX this summer on the Falcon 9 Payload Structural Accomodations team
- Senior advising member of Michigan Mars Rover Team
- Work in the WC as a CNC/Manual Mill trainer

Fun Facts

- Grew up in San Jose, CA
- Cute dog named Tess!
- I'm my elementary school's 2012 hula hooping champion



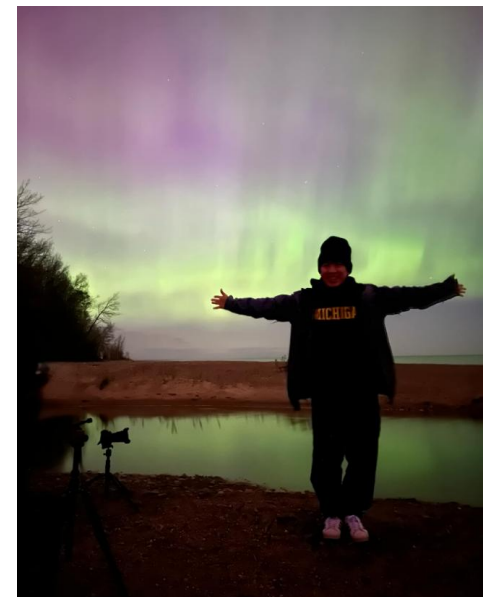
Lawrence Chen

Background

- BS and MS in mechanical engineering at National Taiwan University
- Worked on VR headsets at HTC Vive
- Currently developing microfluidics devices

Fun Facts

- Is an aurora chaser
- Has groundhogs as neighbors
 - They don't pay rent, but they do pose for photos.



Haoran Shi

Background

- Fourth year Ph.D. student in ME in Manufacturing
- Working on simulation of the metal additive manufacturing process
- ME M.S. University of Michigan (2020-2022)
- ME B.S. Rose-Hulman Institute of Technology (2016-2020)

Fun Facts

- Done oil change on an airplane, but not on my car



James Lorenz

Background

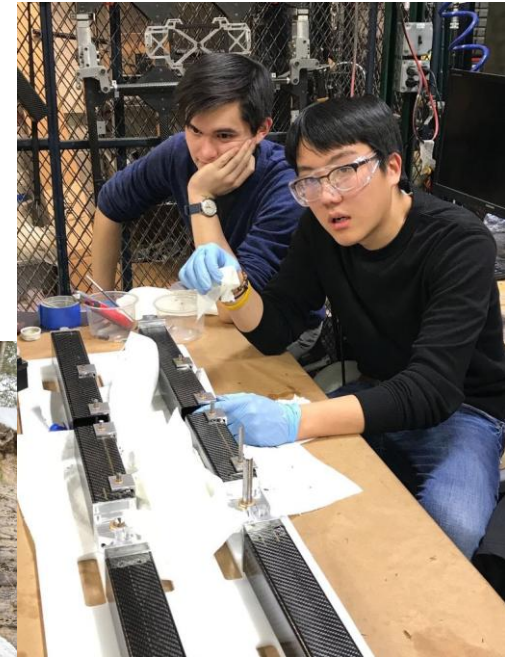
- MechE PhD candidate in mechatronics and robotic additive manufacturing
- Studied MechE and business administration at UM for undergrad
- Worked on MROver and did research in soft-material additive manufacturing in undergrad
- Went to high school in Saratoga, CA

Fun Facts

- Has a trivia fact about every Pokémon (knowledge limited to Gen 1-5)
- Likes to rock climb, cook, bike, and play piano
- Knows all the best restaurants in Ann Arbor. Prove me wrong!



Me enjoying life



Also me enjoying life
(Time: 2:33am)



Moeen Radgolchin

Background/Research area:

- Bachelor and Master: Mechanical Engineering, Solid Mechanics
- Worked in industry for 5 years
- PhD candidate at PSDL: design and analysis of flexure mechanisms for high-precision motion applications



Fun facts:

- Used to play soccer professionally in high-school (still play!!)
- Love music



Nana Porter-Honicky

Background

- B.S. in Mechanical Engineering from UC Berkeley
- Worked on instrumentation for a dark matter detector for two years
- Currently in the Neurobionics lab working on computational biomechanics and rehabilitation research

Fun Facts

- I worked as a pastry chef in a restaurant for two years
- I love swimming, reading, baking, quilting, D&D, and my cat Clytemnestra



Don Wirkner



ME Instructional Lab Manager
(Machine Shop and Mechatronics Lab)

dwirkner@umich.edu

Joined the University of Michigan in 2019

Background:

- Michigan Tech ME degrees
- 16 years in design, vehicle development and testing at General Motors
- Formula SAE – 4 years



Fun Facts:

- Lives in Dexter with wife and two boys
- Enjoys camping and hiking
- Have a home hobby auto/machine shop
- Enjoys motorcycles and motorized toys



Jon Yenkel

Mechanical Engineering Technician

jyenkel@umich.edu



Background:

- Been working at UofM since 2017.
Huf North America
- Automotive supplier that makes door handles and key fobs.
- 3D printed and machined prototype parts.

Department of Boost

- CAD design for aftermarket supercharger systems for Ford Mustangs.

Cablearm Clamshell

- CAD design for dredging buckets.



Fun Facts:

- Born in Michigan
- Play paintball
- RC cars
- Started taking a foundry / metal casting class.



Mike Klein

Background

- Engineering Technician. Undergrad machine shop
- **S** MSU Grad.
- Live in Dexter
- Wife and 3 children
- Worked as a manufacturing engineer for the orthopedic industry building knee, hip and spine implants.



Fun Facts

- Like to fly fish and now live on the Huron River.
- Enjoy collecting and rebuilding vintage stereo equipment.



Kemal Duran

Background:

- Completed UG from the University of Michigan (Dearborn)
- Completed an MS from Wayne State University
- Has worked at several different automotive Companies before UofM

Fun Facts:

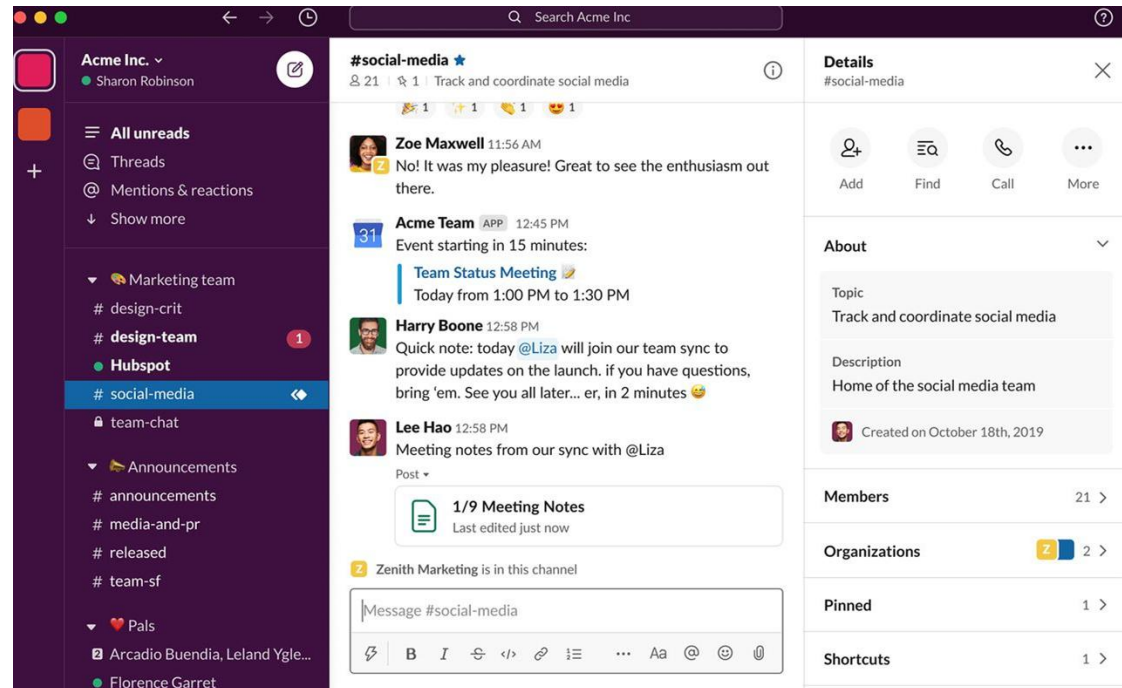
- Avid Dungeons and Dragons Player/ Dice hoarder
- Fluent in Turkish (Parents are from there)
- Enjoys Baking during free time



Communication Policy

- Most communication with the instructors and/or GSIs on course related material should happen via the Slack Workspace.
 - If you have a question, check Slack to see if your question is already answered. If not, post a new question. The instructors will respond within 24 hours.

- You can contact the instructors via email on matters that are personal in nature.
- Once a week we will send out an announcement through Canvas to the full class.



Machine Shop

- If you have not done so already, you must take the **five online machine shop training modules** on the **ME Manufacturing Labs Canvas site**, by September 5th.
 - The five modules are: General Workshop Rules and Information, Cutting/Drilling, Lathe, Mill, and Precision Measurement
 - Send email to ME350F25instructors@umich.edu to request an invitation if you do not already have access to this site.
- Students must complete the **Mill and Lathe Exercises** before September 30th if they have not already completed these exercises in ME250.

All materials for the Mill and Lathe exercises will be provided in the machine shop.

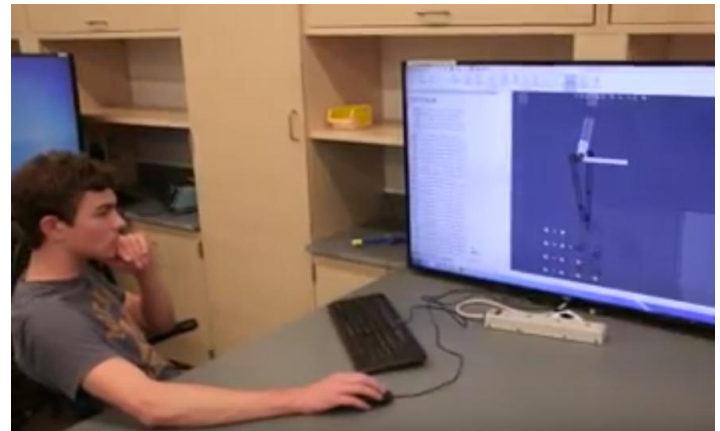


Machine Shop (continued)



- **After September 1st, advance registration is required to use the mill and lathe machines.** Go to the “Undergraduate Machine Shop” link on the M.E. Student Intranet to sign up for a time slot on a machine.
- **Missing a reserved time slot in the machine shop will be considered a serious offense.** Teams that need to miss their time slot should cancel the reservation (on the same website that they used to make the reservation.) Any team that misses a reserved time slot without cancelling it will lose machine shop privileges until the end of the following day.
- Machine shop is open **8:30am-8:30pm M, Tu, Th, Fr, and 8:30am-4:30pm W**

Prerequisites for ME350



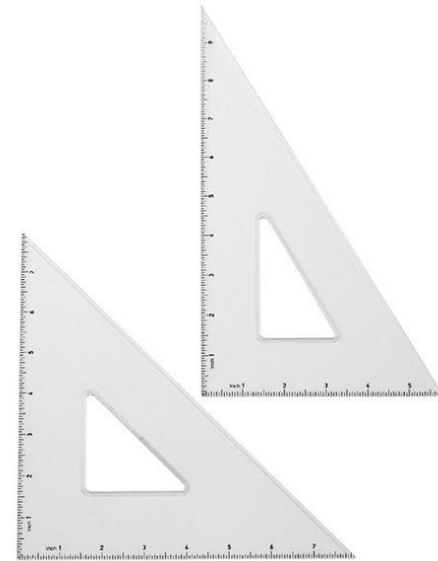
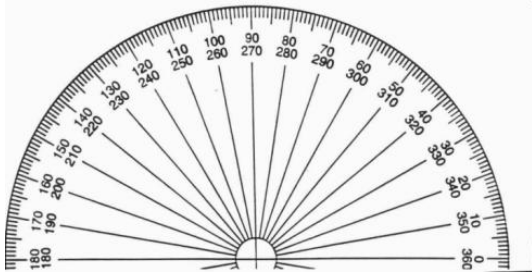
You are expected to have:

- (i) a basic working knowledge of elementary mechanics: statics (e.g. ME211) and dynamics (ME240)
- (ii) basic machine shop skills (i.e., lathe, mill, drill, etc.)
- (iii) working knowledge of SolidWorks

Optional Textbook! (for reference only; see the syllabus)

But there are some **Required** tools

Two plastic triangles, a ruler, a protractor, a compass, and Safety glasses



Homework

- Assigned on Canvas approximately a week before it is due.
- We encourage you to work together on homework in this class. Each person must do their own work.
- We have an open solution policy during office hours, which means you can see the **final answer** to the problem – **but you need to have already worked the problem before coming to office hours.**
- You will submit the HW to Gradescope, accessed from Canvas.
- You are not allowed to possess, look at, or use a solution (paper or electronic) prepared in prior years.



Late Policies

- **Homework:**

- You get a total of **two** late-days, each granting you a 24h extension beyond the deadline (weekend days are counted too), which you can use at your own discretion for homework over the course of the semester.

- **Project Deliverables:**

- For project deliverables such as the **Design Reports**, you will be given a **grace period of 60 minutes** to accommodate for technical issues during submission.
- For more serious issues with deliverables, you can ask for a 24h extension in exchange for a maximum penalty of 25% on that assignment, per the discretion of your lab instructor.

- **Design Lab Assignments --** A follow-up assignment will be associated with each Design Lab session, to be turned in at the start of the following session:

- Similar to the HW policy, but with a total of **four** late-days

Exams



- There will be **two exams** given:
 - one midterm (**in-person, during regular lecture hours**, currently scheduled for **September 30th**) and
 - one final exam (**in-person, on December 11th from 4:00pm-6:00pm**).
- For each exam the students are expected to abide by the honor code and are required to sign the **Honor Pledge** in order for the exam to be graded.
- Make-up exams may be offered only in exceptionally rare cases. If you have extenuating circumstances, you need to contact the instructors prior to the exam. Once the exam has started, no make-up exams will be granted and the missed exam will receive a zero score.
- More information will be provided one week before the exam.

Design Labs

- 2 hours long
- Format varies for each:
 - Software training
 - Projects
 - Design
 - Analysis
 - Build
 - Test



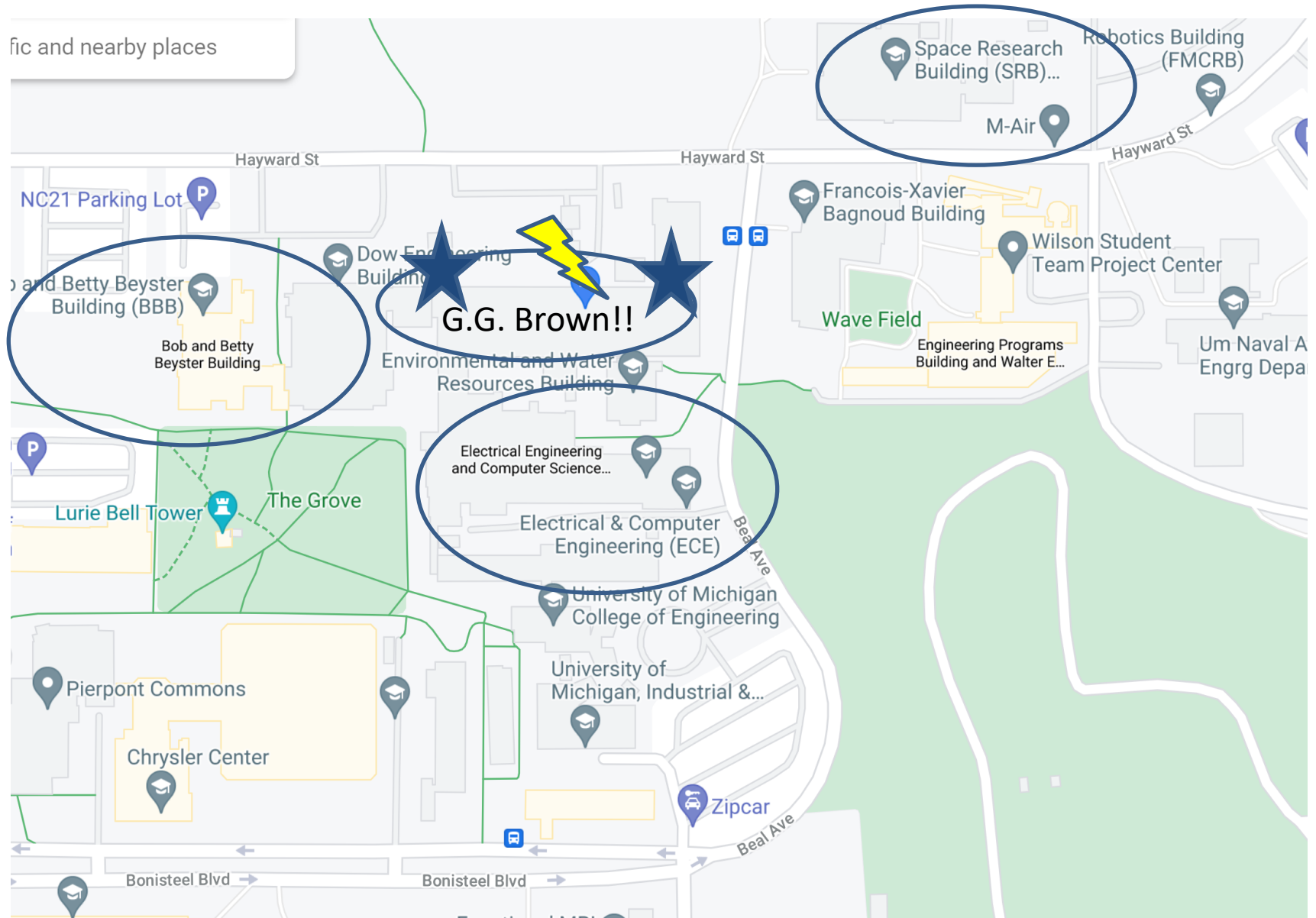
ME350 Lab Sections

Section	Day	Time	Instructor
002	Monday	12:30pm - 2:30pm	Haoran
003	Friday	1:30pm - 3:30pm	Chloe
004	Friday	3:30pm - 5:30pm	Nana
005	Thursday	3:30pm - 5:30pm	Lawrence
006	Monday	10:30am - 12:30pm	Moeen
007	Monday	6:30pm - 8:30pm	James

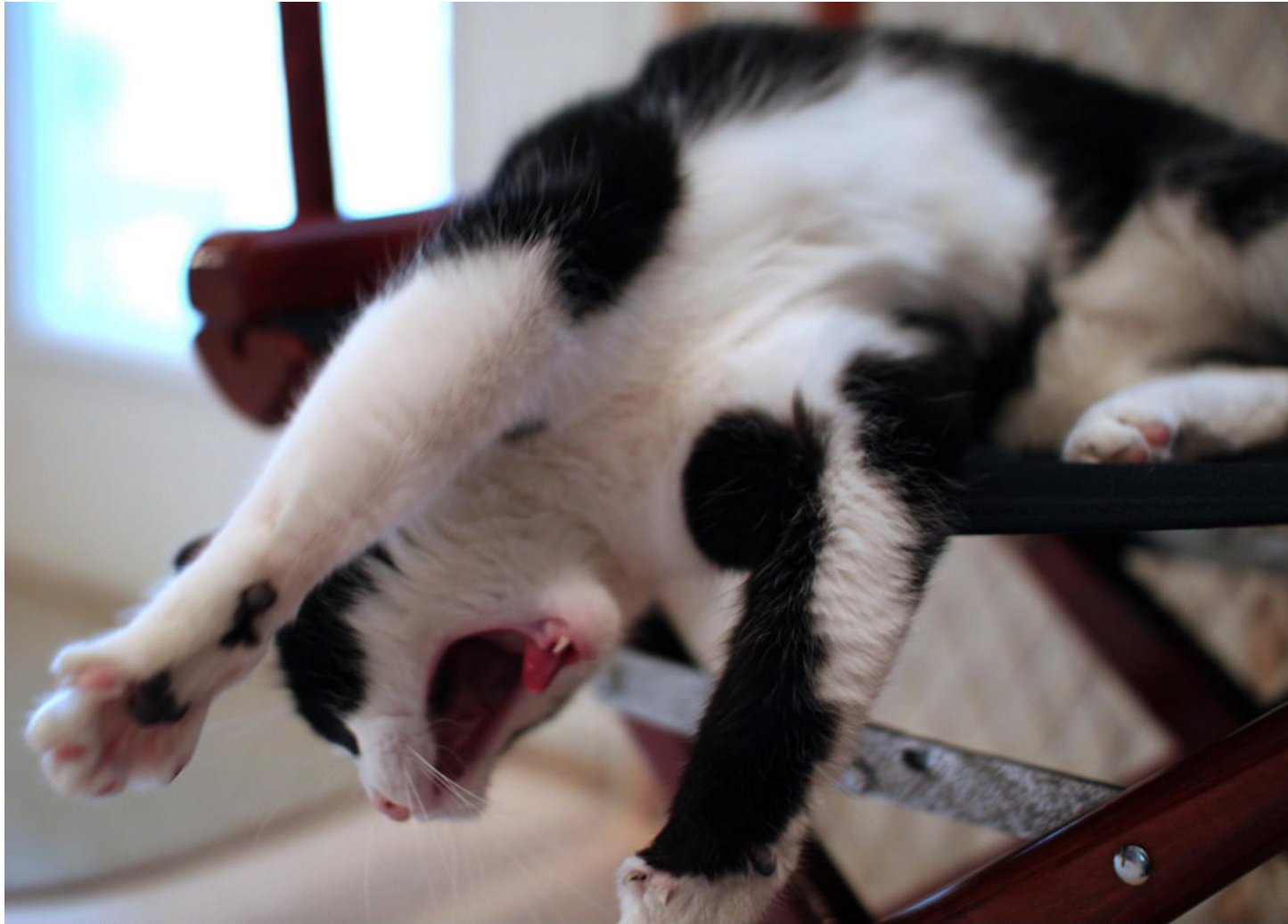
**Lab Sessions will be held in the Mechatronics Lab, room 1345 GGB.
Please bring a laptop if you have one.**

**Also please be able to use SolidWorks on your laptop –
see the SolidWorks Access Guide on the Canvas homepage.**

CAEN Lab Locations



STRETCH!



Grading

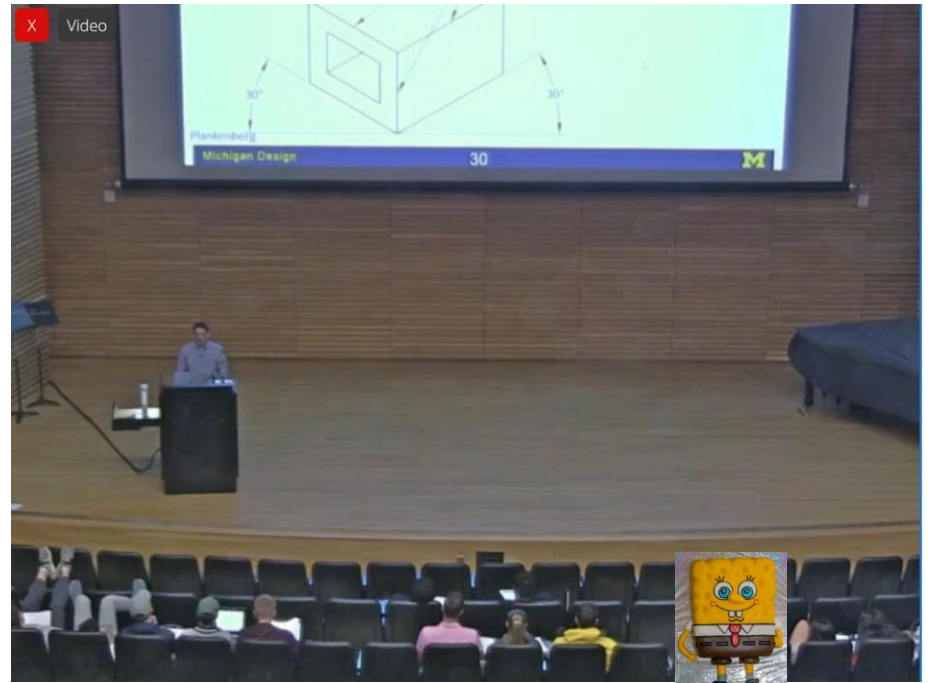
- The numerical average of the class will be set to the historical average for the course. The historical average grade for ME350 is B/B+.
- The total percentage will be broken down as follows:

Project:	50%
Midterm Exam :	15%
Final Exam :	20%
Homework :	<u>15%</u>
Total	100%
- The Peer Evaluations can add or subtract up to 10 percentage points of an individual student's project grade, so they can affect your final grade by up to +/- 5%.
- Project activities will be submitted and scored as a team grade. The exams, homework, and peer evaluation will be scored on an individual basis.

Recording of Lectures

- Course lectures will be audio/video recorded and made available to other students in this course. As part of your participation in this course, you may be recorded.

If you do not wish to be recorded, please contact me350F25instructors@umich.edu during the first week of class to discuss alternative arrangements.



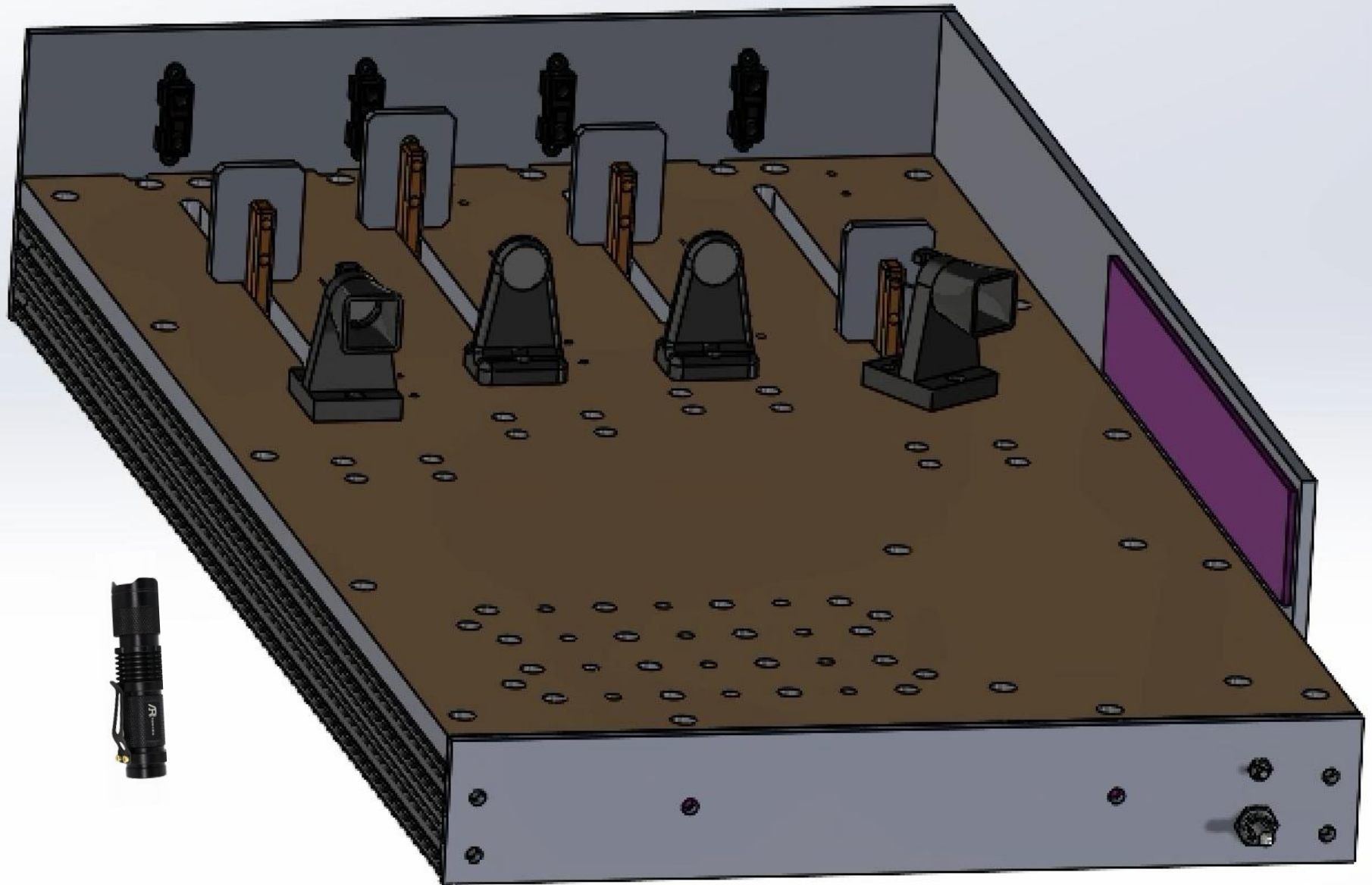
- Students may not record or distribute any class activity without written permission from the instructor.

Design Project



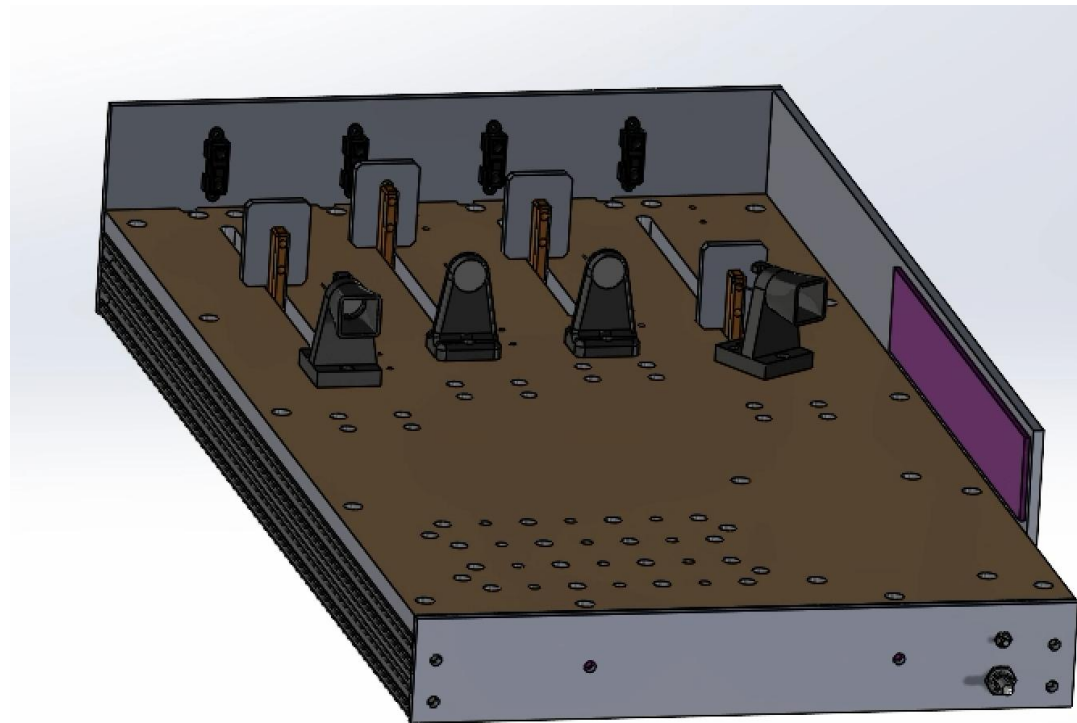
- **Details will be discussed during the first lab session.**
- Students will work in teams to design and build the project. Team assignments will be made by the GSIs and announced during your second lab. All team members must be in the same lab section. Students will be instructed on how to submit their team preferences to their GSI. (**Team Preference Form**)
- There are **two gate reviews** (each of which also has a **design report**) and a **final report** for the project.
- A kit of materials will be provided, and you will also be allowed to purchase other materials.
- Design and machining occurs early; mechatronic components are added later.

Design Project



Design Project

- Goal:
Use model-based design to develop a prototype of a mechatronic system that will detect which targets are active and shine a flashlight on them.
- Aspects:
 - Model-based design
 - 4-bar linkage mechanisms
 - Machine elements
 - Mechatronics (sensors, motors, controllers)
 - Software tools
 - Fabrication
 - Technical report writing and communication
 - Teamwork



Course Schedule

(on the Canvas Homepage)

		F25 Calendar View				Blue = Thursday Lab Purple = Friday Labs Green = Monday Labs		
Week		Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
			Aug 25	Aug 26	Aug 27	Aug 28	Aug 29	Aug 30
1	Lecture			Lecture 1 - Course Overview and Logistics (ALL)		Lecture 2 - Mechanisms: Kinematic Diagrams and Mobility (DB)		
	Homework					HW1 Assigned		
	Lab		Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial			Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial	Lab 1 - Project Overview, SolidWorks Graphical Linkage Synthesis and Sketch Relations Tutorial	
	Project Timeline		Classes Begin - FA25					
September		Aug 31	Sep 1	Sep 2	Sep 3	Sep 4	Sep 5	Sep 6
2	Lecture		Labor Day	Lecture 3 - Mechanisms: Kinematic Inversions, Instant Centers (DB)		Lecture 4 - Mechanisms: Mechanical Advantage and Graphical Synthesis (DB)		
	Homework							
	Lab				Team Preferences Form is Due	Lab 2 - ADAMS Tutorial Lab Assignment #1 Due - Sketch Blocks	Lab 2 - ADAMS Tutorial Lab Assignment #1 Due - Sketch Blocks	

Machine Design

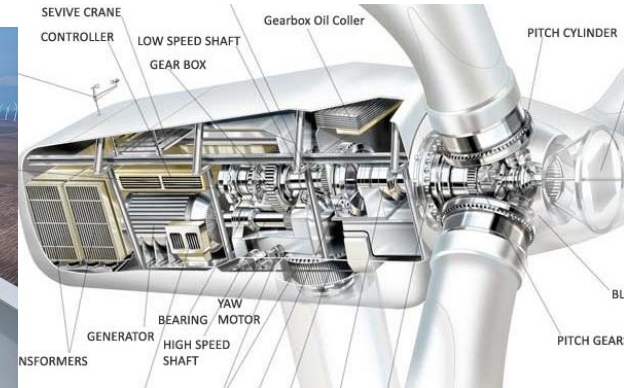
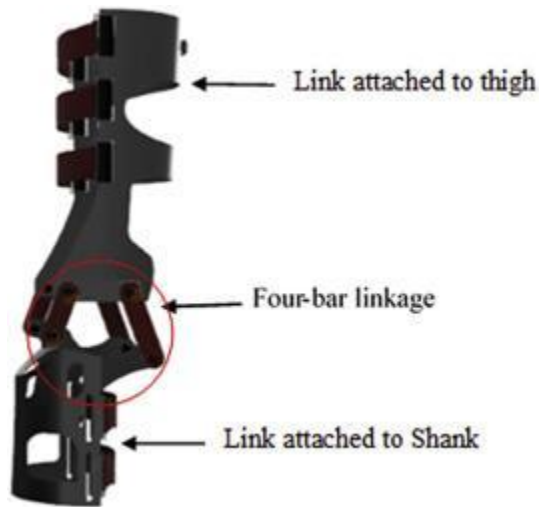
20 years ago



D. Brei

Machine Design

Today...



D. Brei



ME350 PUZZLE

What do the following items have in common?

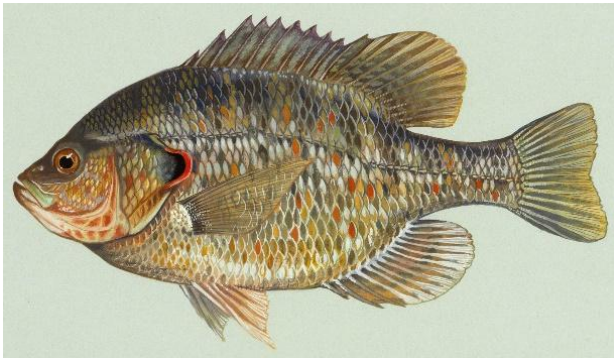
1) Vise-grip
Locking Pliers



2) Umbrella



3) Opercular Jaw



4) Automotive Suspension



ME350 PUZZLE

What do the following items have in common?

Answer: They all have linkages

1) Vise-grip
Locking Pliers



2) Umbrella

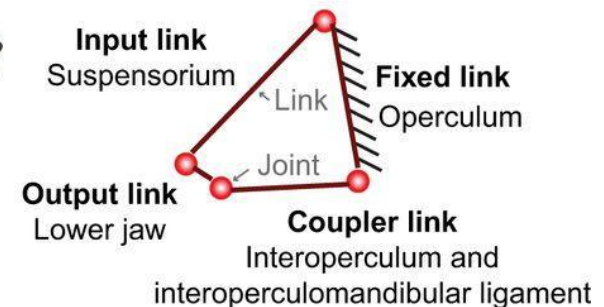
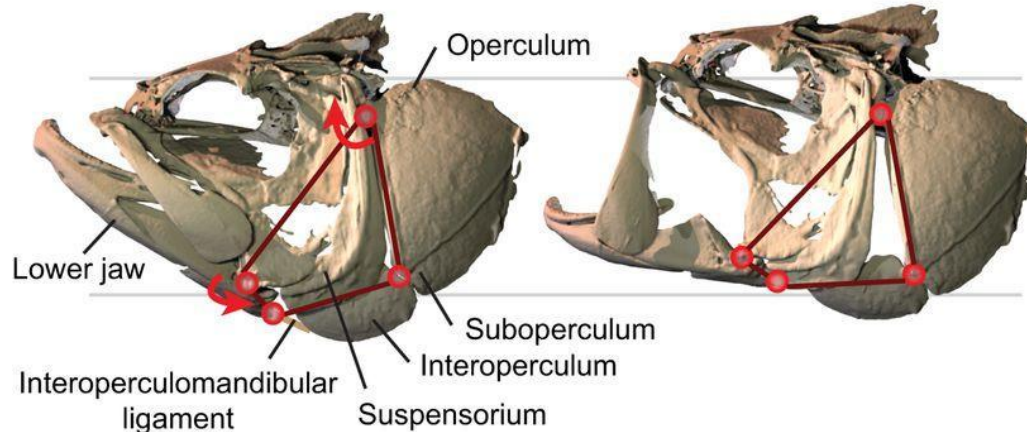


3) Opercular Jaw

A

B

C



Lecture Topics

Mechanisms

Linkages
Kinematic Diagrams
Mobility
Instant Centers
Mechanical Advantage
Graphical Synthesis
Analytical Synthesis

Machine Elements

Fasteners
Introduction to Transmissions
Transmission Ratios
Forces in Gears
Failure of Gears
Bearings

Mechatronics

Introduction to Mechatronics
Motors and Actuators
Analog Sensors
Digital Sensors
Signals
Control

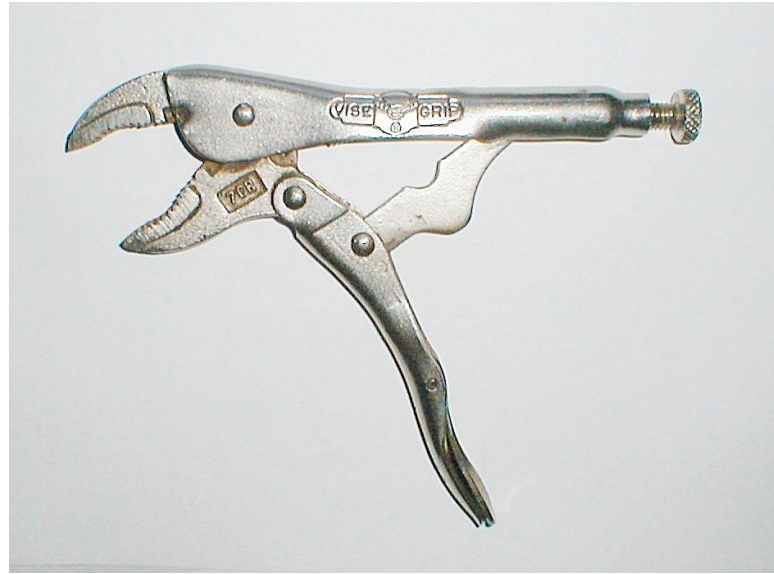
Modern Machine Design

Links, Joints, and Linkages

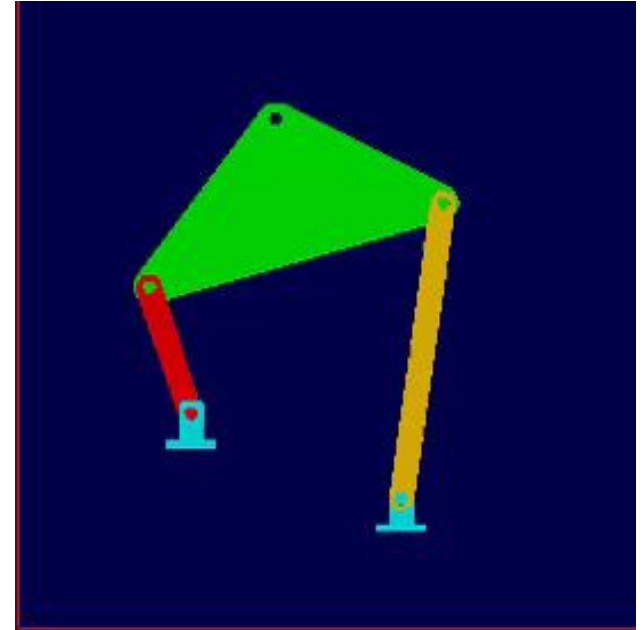
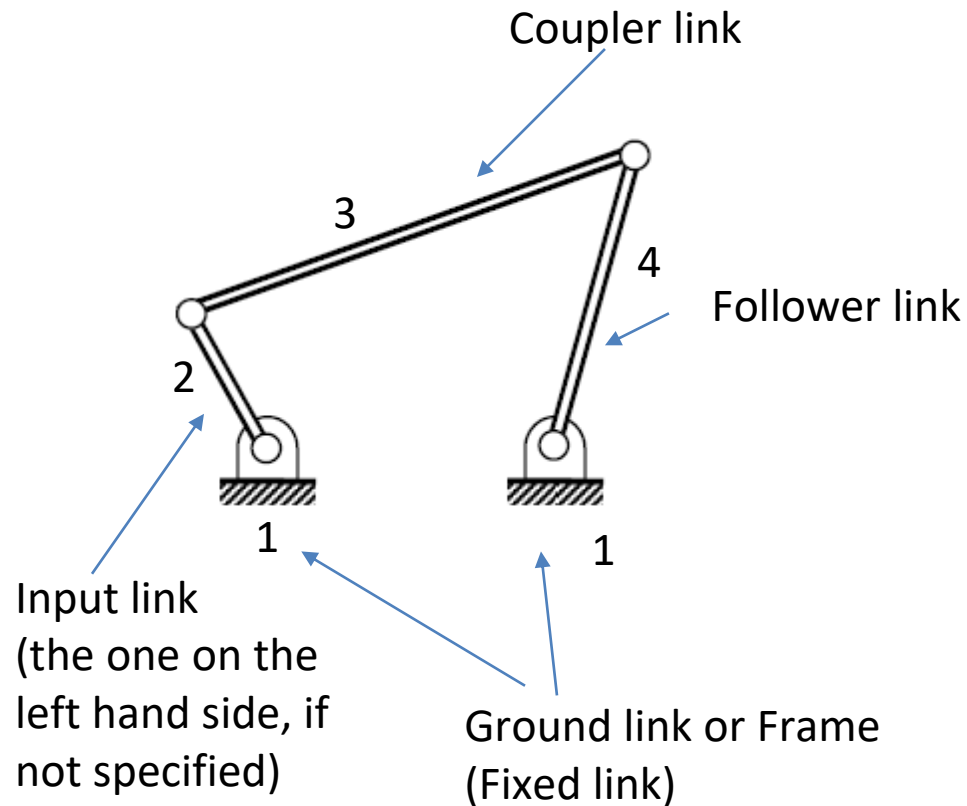
Link: a rigid body with at least two joints

Joint: the connection between two links

Linkage: an assembly of links connected by joints



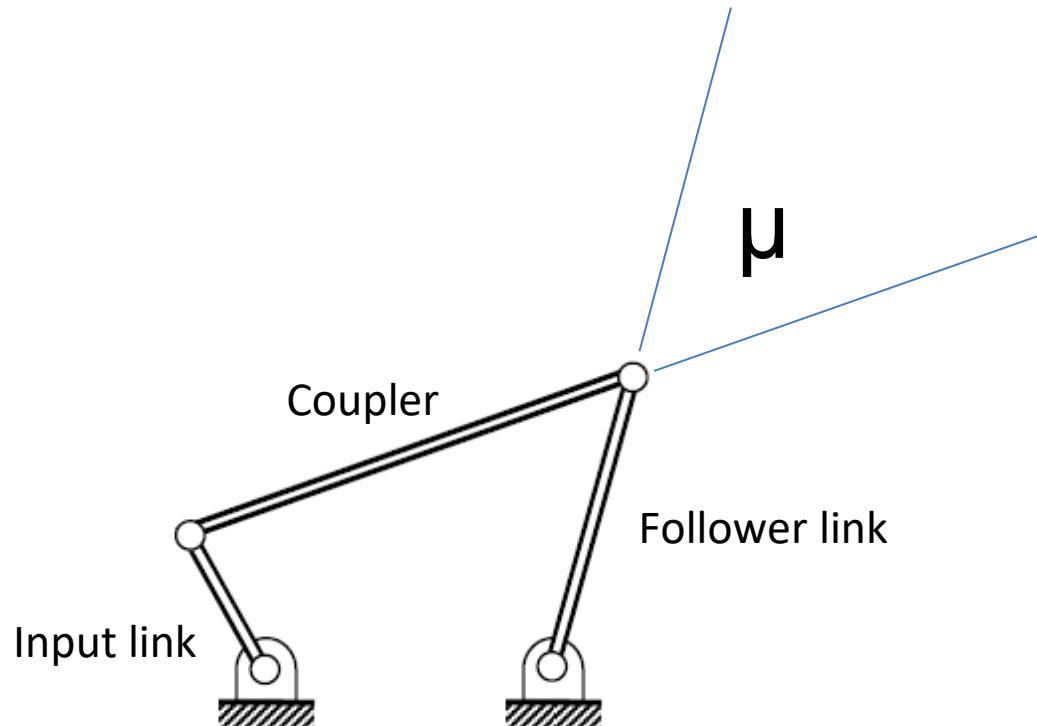
Terminology of a Four-Bar Linkage



Always label ground link as "1"

Transmission Angle, μ

The transmission angle is a means of determining the effectiveness of transferring motion and force to the output link.



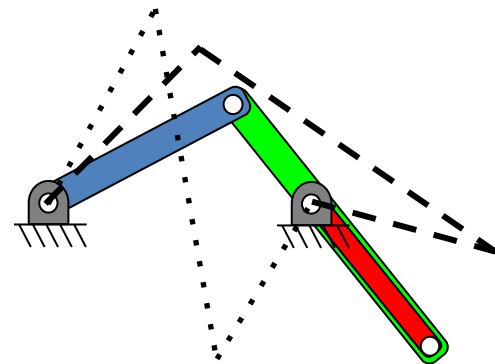
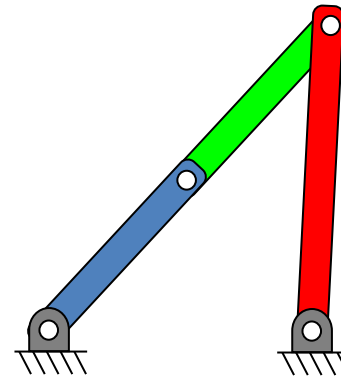
It is defined as the acute angle between the coupler and the follower.

Poor Transmission Angle

- Ideal range of transmission angle
$$\mu \geq 30^\circ$$
$$(\mu \leq 150^\circ)$$
- The following may occur if transmission angle is poor:
 - Mechanism binds (no motion results)
 - Joints fail (pieces go flying)
 - Wrong motion (can't control which way it comes out of the toggle position)

Toggle positions

- Input link collinear with coupler
 - Infinite Mechanical Advantage (theoretically)
 - μ not necessarily = 0
- Coupler collinear with output link (follower)
 - “Change point” (dead center)
 - $\mu = 0$
 - Can’t control which way it comes out of the toggle position



Machines are composed of Mechanisms

A **mechanism** is a device for transferring motion and/or force from a source to an output. In a mechanism, at least one link has been grounded, or attached to the frame of reference (which itself may be in motion.)

Landing Gear Mechanism



When it works...

<http://www.youtube.com/watch?v=aTjUFQvstjU>

<https://www.youtube.com/watch?v=xkp3XRlqAy0>

When it doesn't work...

<http://www.youtube.com/watch?v=ANLgWY6I9uk>